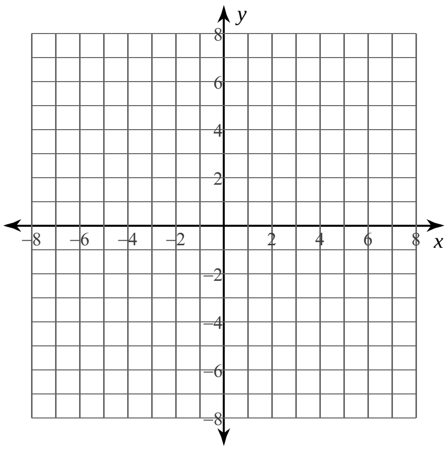
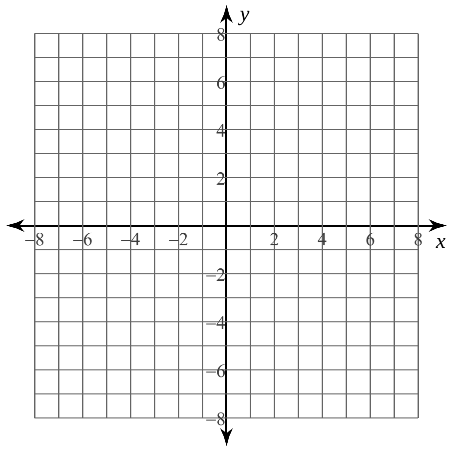
5) $f(x) = -x^3 + 6x^2 - 12x + 8$



6) $f(x) = x^3 - 2x^2$



**Describe the end behavior of each function.**

7) $f(x) = -x^5 + 2x^3 - x + 1$

8) $f(x) = 2x^2 - 4x - 3$

9) $f(x) = x^4 - 2x^2 - x + 1$

10) $f(x) = -x^3 - 9x^2 - 24x - 20$

11) $f(x) = -x^5 + 3x^3 + 1$

12) $f(x) = x^2 + 6x + 6$

**Critical thinking questions:**

13) Write a polynomial function $f$ with the following properties:
(a) Zeros at 1, 2, and 3
(b) $f(x) \le 0$ for all values of $x$
(c) Degree greater than 1

14) Write a polynomial function $g$ with degree greater than one that passes through the points $(0, 1)$, $(1, 1)$, and $(2, 1)$.